

Programme	B. Sc. Computer Science				
Course Title	Full Stack Web Development				
Type of Course	Major				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Fundamental of Web Pages and web servers 2. Basics of HTML				
Course Summary	This course provides the ideas, techniques, and applications for efficient Web Development. The advanced industry demand and emerging trends are covered in this syllabus.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the concepts to create responsive web pages using HTML and CSS	U	C	Instructor-created exams / Quiz
CO2	Familiarization with Client-side Scripting using JavaScript	U	C	Practical Assignment / Observation of Practical Skills
CO3	Understand Node.JS and equip learners with a comprehensive understanding of NodeJS and its functionalities.	U	F	Seminar Presentation / Group Tutorial Work/ Viva Voce

CO4	Understanding and building interactive web pages using React JS.	U	P	Instructor-created exams / Home Assignments
CO5	Familiarization with SQL and NoSQL	Ap	P	Writing assignments/ Instructor-created exams/ practicals
CO6	Explore MongoDB and Develop real-world web applications using various technologies learned in the course	Ap	P	Case Study/ mini Project/ practicals
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45+30)	Marks (70)
I	HTML & CSS		9	12
	1	Introduction to HTML5 Tags, Attribute and Elements Doctype Element, Comments	2	2
	2	Semantic tags Headings, Paragraphs, and Formatting Text Lists, Links, Images-	1	2
	3	Forms and Tables Introduction CSS Applying CSS to HTML.	2	2
	4	Selectors, Properties and Values CSS Colors and Backgrounds CSS Box Model	3	5

	5	CSS Margins, Padding, and Borders CSS Text and Font Properties Webpage Layout Responsive web design	1	1
II	JavaScript & Node.JS		11	18
	6	Introduction to JavaScript Applying JavaScript (internal and external) Understanding JS Syntax	1	2
	7	Introduction to Document and Window Object Variables and Operators Data Types and Num Type Conversion	1	2
	8	Math and String Manipulation Objects and Arrays Date and Time Conditional Statements	2	3
	9	Switch Case Looping in JS Functions	2	2
	10	Node.JS Overview Node.JS - Basics and Setup Node.JS Console Node.JS Command Utilities Node.JS Modules	3	3
	11	Node.JS Concepts Node.JS Events Node.JS with Express js Node.JS Database Access	2	3
III	React.JS		12	20
	12	Introduction Templating using JSX	2	3
	13	Components, State and Props Lifecycle of Components Rendering List and Portals	3	3
	14	Redux and Redux Saga Immutable.js Service Side Rendering	2	3
	15	Unit Testing	2	3
	16	Webpack	3	3
IV	MongoDB		13	20
	17	SQL and NoSQL Concepts	3	4
	18	Create and Manage MongoDB	2	3
	19	Migration of Data into MongoDB	1	3
	20	MongoDB with PHP	1	3
	21	MongoDB with NodeJS.	2	4
	22	Services Offered by MongoDB	3	3
V	Practical Implementations of Full Stack Web Development		30	
	1	Webpage Development using HTML And CSS	25	

		• Webpage Development using Javascript & Node.JS • Webpage Development using React.JS With Backend MongoDB		
	2	Case Study/ Project	5	

References Books

1. Hawramani, Ikram. HTML, CSS and JavaScript for Complete Beginners: A Step by Step Guide to Learning HTML5, CSS3 and the JavaScript Programming Language. United States, Amazon Digital Services LLC - KDP Print US, 2018.
2. Soni, Ravi Kant. Full Stack AngularJS for Java Developers: Build a Full-Featured Web Application from Scratch Using AngularJS with Spring RESTful. United States, Apress, 2017.
3. Northwood, Chris. The Full Stack Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Full Stack Web Developer. Germany, Apress, 2018.
4. Sharma, Aneeta. Full-Stack Web Development with Vue. Js and Node: Build Scalable and Powerful Web Apps with Modern Web Stack: MongoDB, Vue, Node. Js, and Express. United Kingdom, Packt Publishing, Limited, 2018.
5. Sharma, Manu. Mongoddb Complete Guide: Develop a Strong Understanding of Administering Mongoddb, Crud Operations, and Mongoddb Commands. India, Bpb Publications, 2021.

Mapping of COs with PSOs and POs :

	PS O1	PS O2	PS O3	PS O4	PS O5	PS O6	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6
CO 1	-	1	3	2	2	3						
CO 2	-	1	3	3	3	2						
CO 3	-	1	3	3	3	2						
CO 4	-	1	3	3	3	2						
CO 5	-	1	3	3	3	2						
CO 6	-	1	3	3	3	2						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment

Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	End Semester Examinations	Practical
CO 1	✓		✓	
CO 2	✓	✓	✓	✓
CO 3	✓	✓	✓	✓
CO 4	✓	✓	✓	
CO 5	✓	✓	✓	✓
CO 6	✓		✓	✓